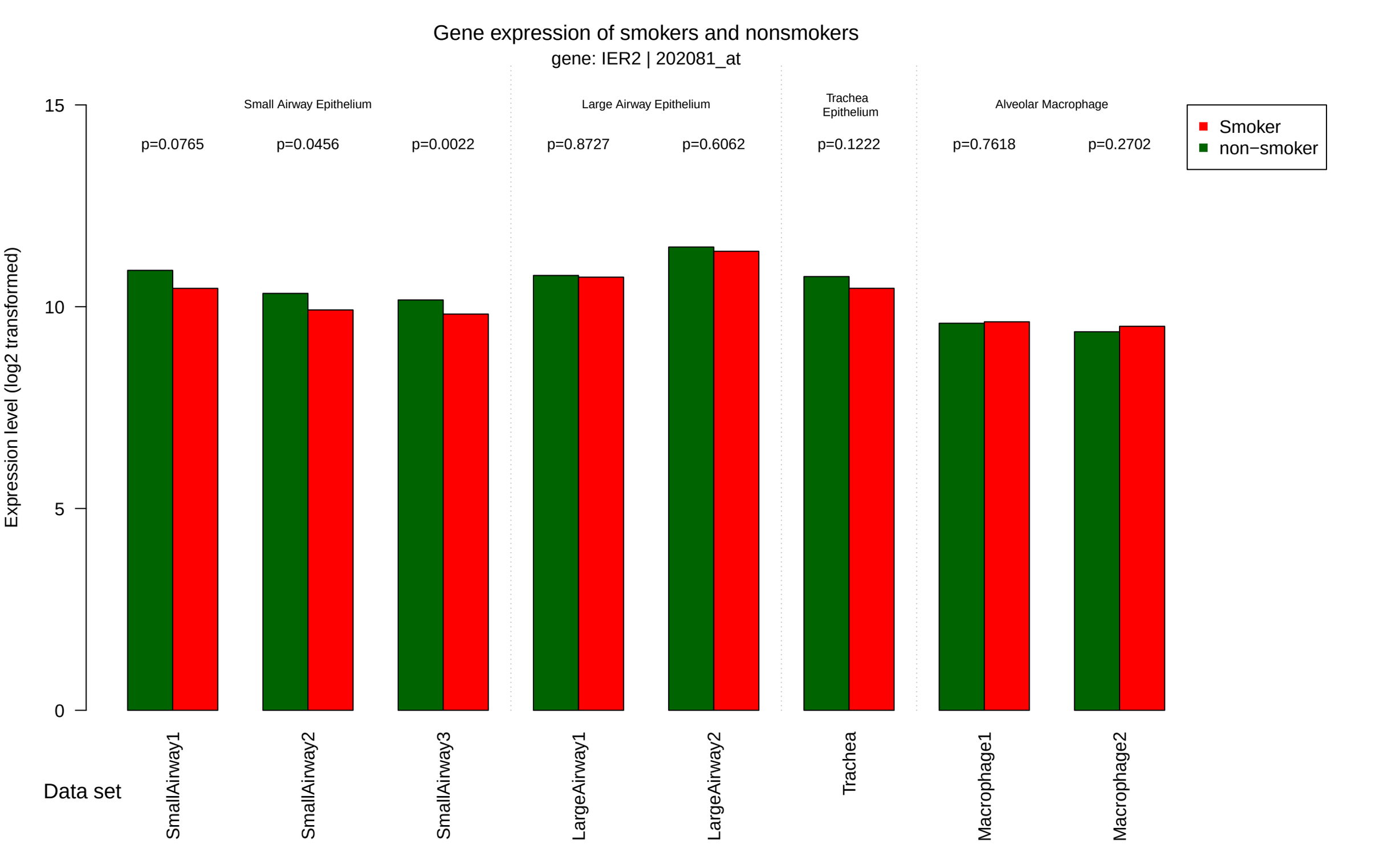
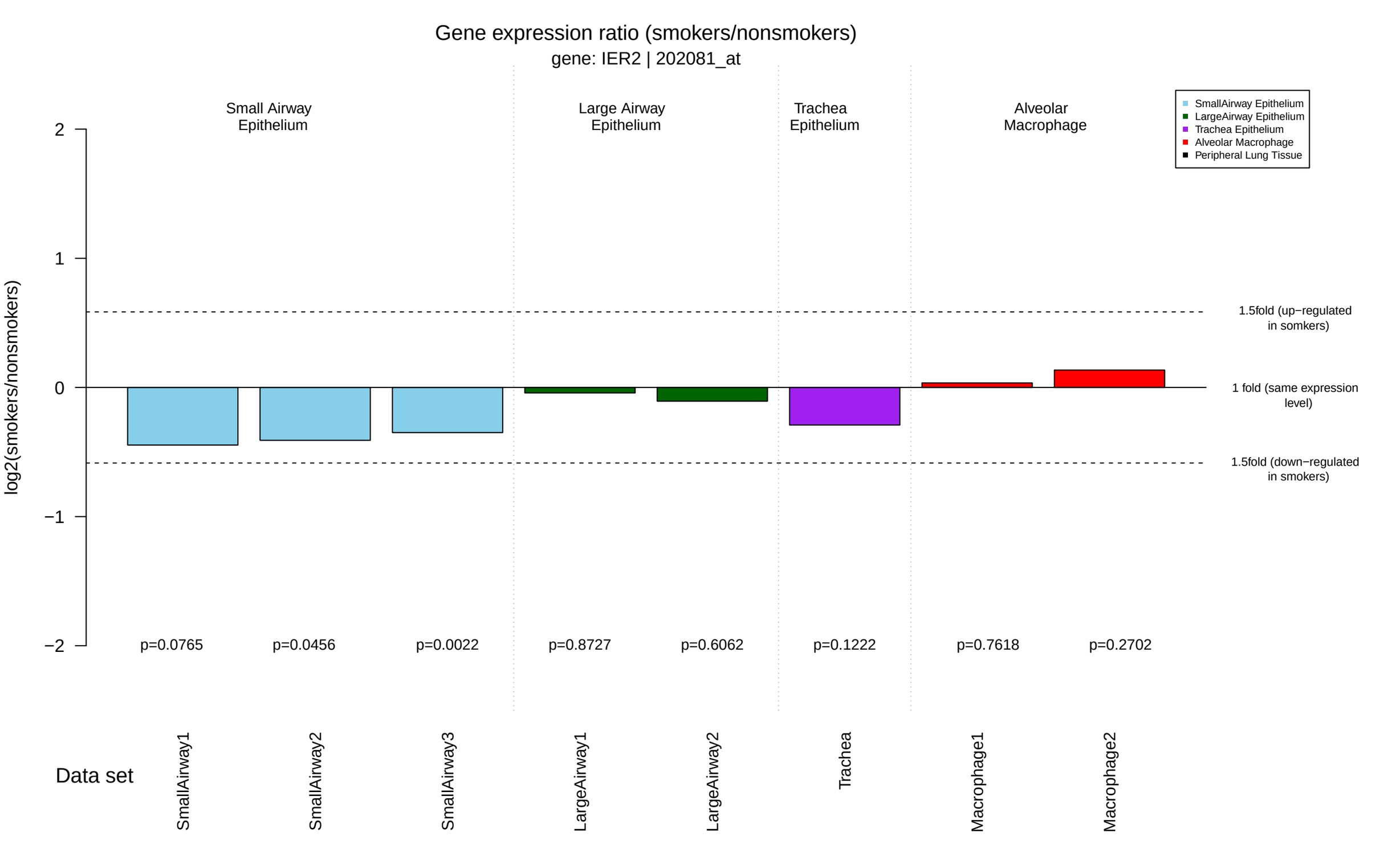
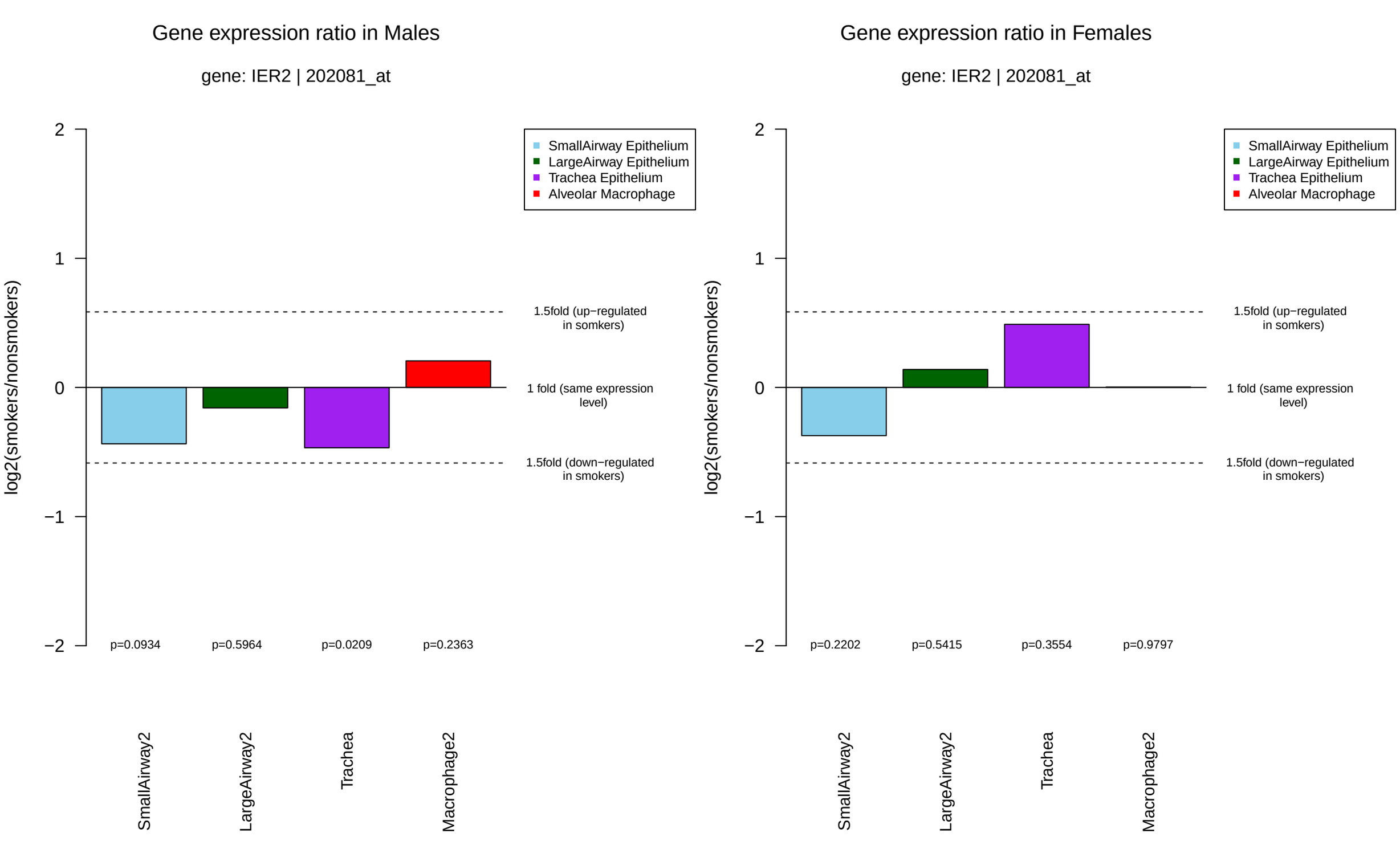
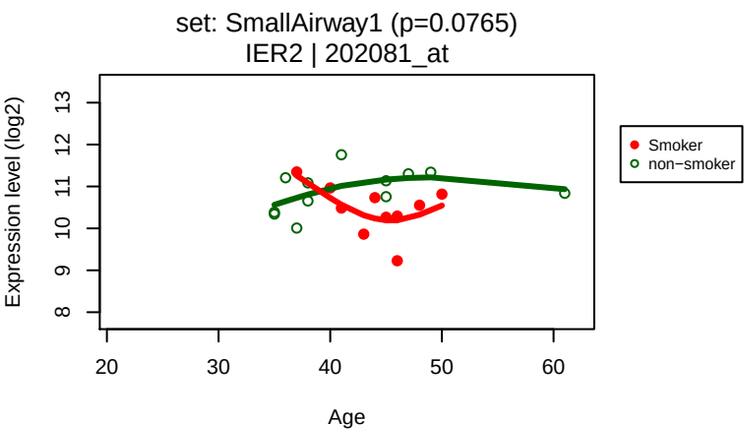
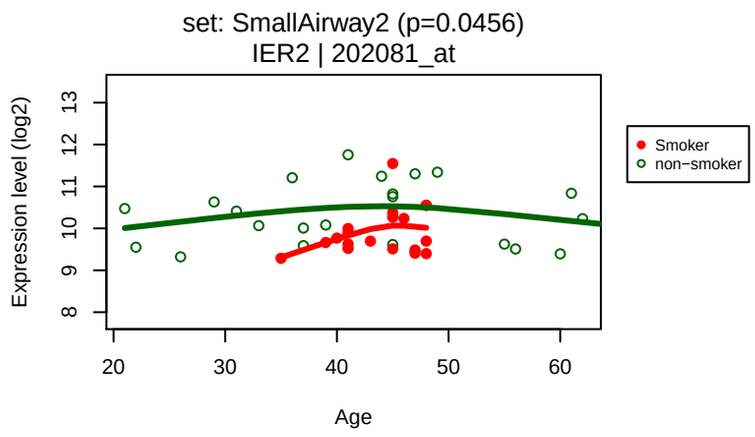
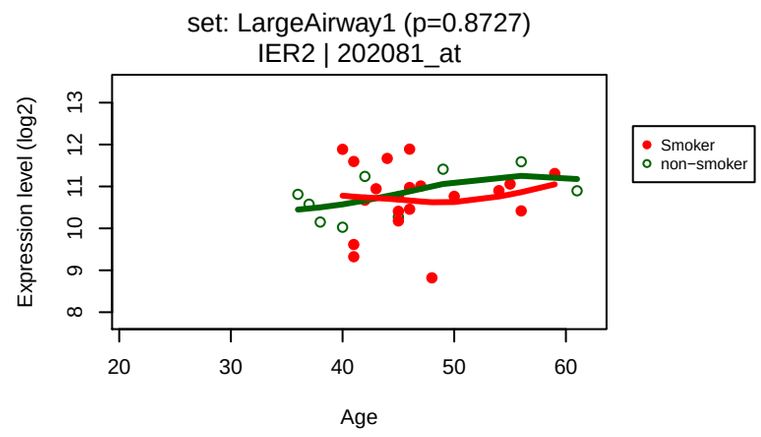

Gene expression of smokers and nonsmokers
gene: IER2 | 202081_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0765
p=0.0456
p=0.0022
p=0.8727
p=0.6062
p=0.1222
p=0.7618
p=0.2702
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

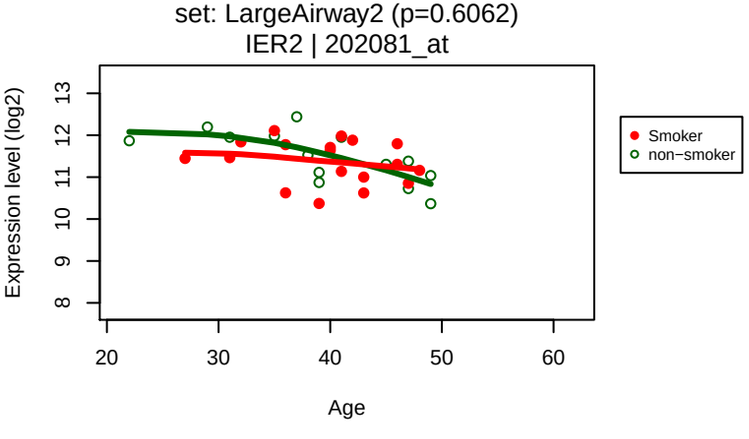

Gene expression ratio (smokers/nonsmokers)
gene: IER2 | 202081_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0765
p=0.0456
p=0.0022
p=0.8727
p=0.6062
p=0.1222
p=0.7618
p=0.2702
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

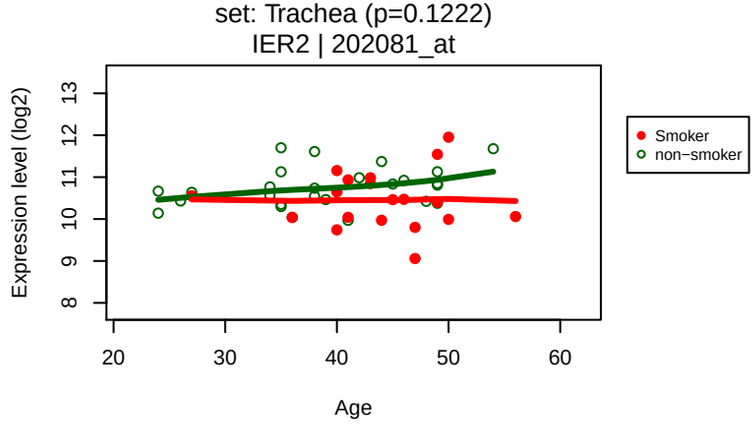

Gene expression ratio in Males
gene: IER2 | 202081_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0934
p=0.5964
p=0.0209
p=0.2363
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IER2 | 202081_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2202
p=0.5415
p=0.3554
p=0.9797
SmallAirway2
LargeAirway2
Trachea
Macrophage2

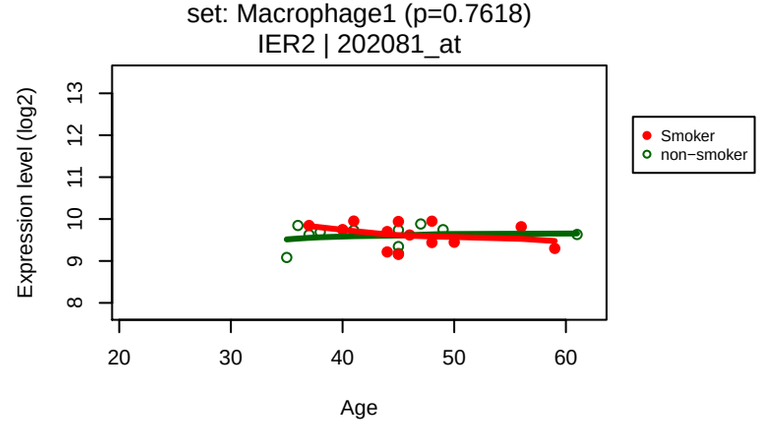

set: SmallAirway1 (p=0.0765)
IER2 | 202081_at
Expression level (log2)
Age
Smoker
non−smoker

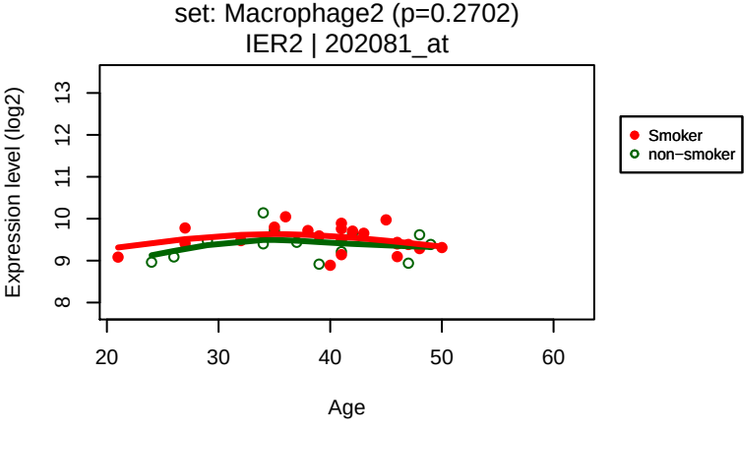

set: SmallAirway2 (p=0.0456)
IER2 | 202081_at
Expression level (log2)
Age
Smoker
non−smoker

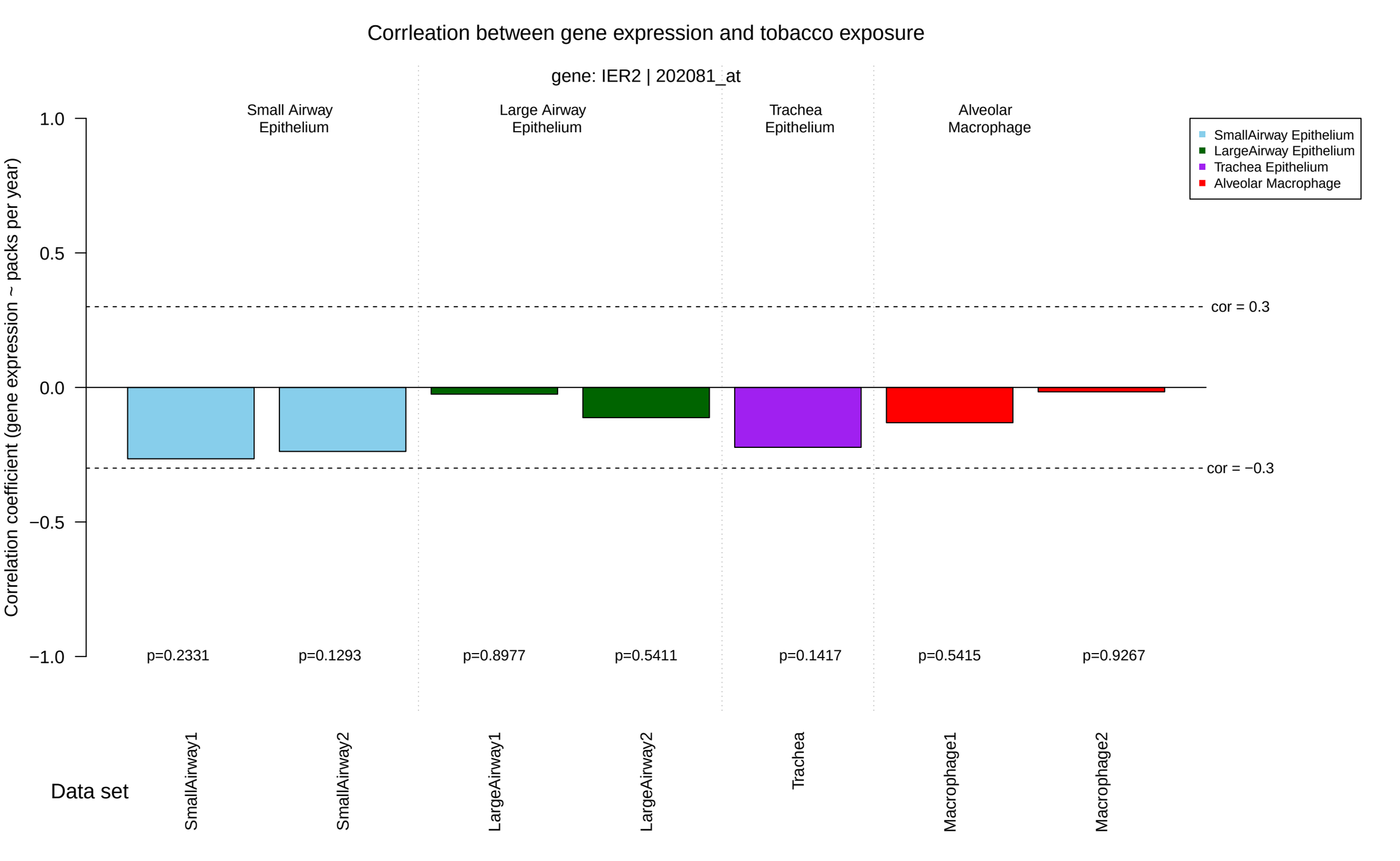

set: LargeAirway1 (p=0.8727)
IER2 | 202081_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6062)
IER2 | 202081_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1222)
IER2 | 202081_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7618)
IER2 | 202081_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2702)
IER2 | 202081_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IER2 | 202081_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2331
p=0.1293
p=0.8977
p=0.5411
p=0.1417
p=0.5415
p=0.9267
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2